

Pranati Modumudi

pranmod01.github.io | (408)-464-8674 | pranati.m@columbia.edu | linkedin.com/in/pranatimodumudi/

EDUCATION

Columbia University

Aug 2025 - May 2027 (Expected)

M.S. in Computer Science, MS Thesis Track in Machine Learning and Computational Neuroscience

Relevant Coursework: Artificial Intelligence, LLM-Based Generative AI, Computation and the Brain, Theoretical Neuroscience

University of California, Berkeley

Aug 2019 - May 2023

B.A. Data Science Honors and B.A. Economics | Regents and Chancellor's Scholar | Berkeley SCET

Relevant Coursework: Data Structures; Linear Algebra and Differential Equations; Probability & Probabilistic Modeling for Data Science; Data Engineering; Natural Language Processing; Game Theory

SKILLS

- ❖ **Programming & Tools:** Python, R, SQL, Bash, Java, JavaScript, Git, Docker, Databricks, Snowflake, Tableau, Streamlit
- ❖ **ML/DL:** PyTorch, TensorFlow/Keras, JAX, Scikit-learn, HuggingFace, AllenNLP, SpaCy, Neural Networks, MLflow, Weights & Biases
- ❖ **LLMs & Generative AI:** Large Language Models, Prompt Engineering, Retrieval-Augmented Generation (RAG), RLHF, Model Fine-Tuning, Model Context Protocol (MCP), Langchain
- ❖ **Data & Research:** Experimental Design, Model Evaluation & Interpretation, Exploratory Data Analysis, Data Wrangling, ETL Pipelines, APIs, Big Data Systems

INDUSTRY EXPERIENCE

Disney Streaming, New York

Data Scientist I

Aug 2023 - May 2025

- ❖ Built end-to-end ML pipelines for time-series and user-level causal inference using Bayesian structural models, enabling real-time monitoring with quantified uncertainty (80-90% confidence intervals) and reducing analysis turnaround from weeks to days.
- ❖ Conducted ~24 causal analyses evaluating pricing and marketing initiatives affecting millions of Disney+ and Hulu subscribers, directly informing executive strategic decisions.
- ❖ Developed reusable Python package for synthetic data generation, continuous treatment models, and time-series forecasting; deployed via PyPi and Jenkins, adopted by five teams.

OctoML, Remote

Data Science Intern

Oct 2022 - Dec 2022

- ❖ Optimized TVM machine learning compilation software by running ~100 experiments on convolution models of varying sizes, tuning hyperparameters to reduce convergence time by ~10%.

Machine Learning Systems Intern

Mar 2022 - Jun 2022

- ❖ Diagnosed causes of performance variance in TVM auto-tuning pipeline as part of the Automation team; [published](#) findings accepted to MLArchSys International Symposium on Computer Architecture 2022.

RESEARCH/PROJECTS

Human Decision Modeling | Laboratory for Intelligent Imaging and Neural Computing, Columbia University

- ❖ Developing real-time EEG and eye-tracking pipelines to extract physiological markers of uncertainty and risk aversion, informing reward shaping for aligned AI agents and adaptive brain-computer interfaces.
- ❖ Built preprocessing pipelines for multi-modal physiological data from decision-making experiments, enabling real-time decoding of human cognitive states under risk and ambiguity.

Multi-Agent SLM Evaluators for AI Slop Detection | AWS Small Language Build Day Hackathon

- ❖ Built an ensemble of four specialized small language models (1-3B parameters) fine-tuned with LoRA on AWS Trainium to detect low-fidelity model outputs, achieving 10× cost reduction vs. large-model inference for real-time content quality monitoring.

AI Capability Terrain Map | Apart Research AI Forecasting Hackathon

- ❖ Developed an interactive 3D early warning system visualizing AI capabilities with real-time Epoch AI benchmark data, logistic growth modeling, and Monte Carlo forecasting across 30+ domains to identify capability trajectories, sinkholes, and emerging risks.

Undergraduate Senior Honors Thesis | Redwood Center for Theoretical Neuroscience UC Berkeley

- ❖ Selected for 15-student cohort; extended hippocampal place cell modeling from rats to bats using complex-valued neural networks to capture spatial coding differences across species under Prof. Friedrich Sommer and Dr. Gautam Agarwal.